# AI for Scientific Writing

## Session 2 · Technical and scientific writing — your topic, your sources

Dr. Leopoldo Rodríguez Salazar

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Tecnológico de Monterrey · Campus Querétaro · Semana Tec

# Yesterday in one minute — and today

**Yesterday**

- LLMs predict the *plausible*, not the *true* ⇒ hallucinations
- temperature, context, grounding/RAG, agents
- six parts of a prompt; check every image
- **The model assists; you are the author.**

**Quick check:** one thing the AI got wrong in Task 1 or 2 — and how you caught it.

**Today (all individual)**

| | |
|---|---|
| 07:10 | Technical and scientific writing |
| 08:10 | Break |
| 08:20 | Activity 4 — define your topic |
| 08:40 | Activity 5 — find your sources |
| 09:20 | Activity 6 — present them (5 %) |
| 09:55 | Tasks 3 and 4 |

# Principles of technical and scientific writing

1. **Clarity** — one idea per sentence, one topic per paragraph.
2. **Precision** — numbers with units and uncertainty; no “very”, “a lot”.
3. **Concision** — cut every word that adds nothing.
4. **Objectivity** — claims supported by data or a cited source.
5. **Reproducibility** — another person could repeat what you did.
6. **Defined terms** — define on first use; spell out acronyms.
7. **Consistent style** — tense, units, citation format.
8. **Honesty** — limitations stated; AI use declared.

Official content 2.1 — Principles of technical and scientific writing.

# Before and after

| Vague | Technical |
|---|---|
| The robot works very well and is much faster than other methods. | The prototype cleaned a $1.6\,\mathrm{m}^2$ panel in 94 s ($n = 10$, SD 6 s), 31 % faster than manual cleaning. |
| AI is revolutionising engineering. | In our survey, 22 of 30 students (73 %) used a generative AI tool weekly for coursework. |
| Many studies show that vibration is important. | Vibration amplitude above 7 mm/s predicted bearing failure within 30 days in two field studies [3], [5]. |

Illustrative numbers. **Rule:** every number comes from your data or a cited source — never from the AI.

# Scope, objective and structure of a report

**Three kinds of article**

| | |
|---|---|
| **Review** | organises what others found; no new data |
| **Research** | your method, your data, your results |
| **Brief summary** | short communication: one result, compact |

**Scope:** what is in and what is out.
**Objective:** one sentence, a verb you can check — *compare*, *measure*, *review*.

**IMRaD — the usual structure**

| | |
|---|---|
| Title | specific, informative, no hype |
| Abstract | 150–250 words: problem, method, result, meaning |
| Introduction | context → gap → objective |
| Methods | what you did, reproducible |
| Results | what you found: figures, tables |
| Discussion | what it means; limitations |
| Conclusions | answer the objective |
| References | every claim traceable |

# Searching and consulting information

**Where**

- **Biblioteca Digital Tec** (Scopus, IEEE Xplore, ScienceDirect)
- **Google Scholar**, Semantic Scholar
- SciELO, Redalyc — research published in Spanish
- standards, patents, official reports

**In several languages:** most engineering research is in English — read it, and cite it.

**How — a search string**

```
("solar panel" OR photovoltaic)
AND (cleaning OR soiling)
AND robot*
```

Quotes = exact phrase · OR = synonyms · AND = must have · * = any ending. Filter by year (last 5–10) and by type (journal, conference).

**AI helps** with synonyms and keywords — **not** as the source of the reference.

# Is this source reliable? — six questions

1. **Does it exist?** DOI opens at doi.org; same title, authors, year.
2. **Who wrote it?** Authors and institution you can find.
3. **Where?** Peer-reviewed journal or conference — not a predatory journal or a content farm.
4. **When?** Recent enough for the topic.
5. **Evidence?** Data and methods, not only opinions.
6. **Does it say that?** You read the abstract *and* the results.

**Peer-reviewed $\neq$ grey literature** (theses, reports, news, blogs). Grey literature can support context — not your main claims.

**Predatory journals:** pay-to-publish, no real review, aggressive emails, promise publication in days.

**An AI answer is not a source.** The source is what the AI points to — after you open it.

# Review, group and summarise — the literature matrix

| # | Reference | Main idea | Method | Key result | For my article | Verified |
|---|---|---|---|---|---|---|
| 1 | Author (year), journal, DOI | what they claim | experiment, simulation, survey, review | the number or finding | supports / contrasts / context | DOI opened |
| 2 | | | | | | |
| 3 | | | | | | |

**Group by theme, not by source:** "two studies measured X [1], [3]; one disagrees [2]".

**Summarise in your words** — ideas, points of view, methods. Quote only when the wording matters.

# Editing text with GenAI — translation, spelling, style

```
Correct spelling and grammar only.
Do not change the meaning, do not add
content, keep technical terms.
List every change you made.
Text: "..."
```

```
Translate into Spanish for an engineering
audience. Keep technical terms; list the
terms you were unsure about.
```

**Fine:** spelling, grammar, readability, translation you check.

**Declare:** restructured paragraphs, rewritten sections, new text.

**Check translations:** technical terms are where tools fail ("stress" $\neq$ "estrés" in mechanics: *esfuerzo*).

**Keep your voice:** accept changes one by one, not all at once.

# Integrating knowledge and analysing results with GenAI

```
Below are the abstracts of my 5 sources,
numbered [1]-[5].
Group them by approach. List where they
agree and where they disagree, citing the
source number for every point.
Use ONLY these abstracts. Do not add
sources. Say if something is not in them.
[1] ...
```

**Good uses:** compare sources you provide · find agreements, contradictions, gaps · suggest how to order the argument · check consistency between your methods and results.

**Then verify:** every "[3] says" — open [3] and check it says that.

**Never:** let it produce your results or your numbers.

# Live demo — from 3 abstracts to an outline

1. Search: one string in Google Scholar → choose 3 papers (DOIs open).
2. TECgpt: paste the 3 abstracts → literature matrix (prompt from the last slide).
3. Check two cells against the papers — find the error, if there is one.
4. Translate one English abstract; spot one technical term the tool got wrong.
5. Ask for an IMRaD outline *from the matrix only*.

The official design asks for a brief live demonstration of the tools for structure, language and sources.

# Activity 4 — define your topic (20 min, individual)

Fill in the **topic card** (handout), then validate it with me:

- **Topic** — specific (not "renewable energy" but "soiling losses in rooftop panels in Querétaro")
- **Question** — what you want to answer
- **Objective** — one checkable verb
- **Type** — review, research, brief summary
- **Audience** — who will read it

**It is the same card RigorIA asks for** in Task 4: topic, objective, article type, audience.

**Too broad?** Ask the AI to propose three narrower versions — you choose.

**Your project counts:** a capstone, lab or competition with your own data is ideal.

# Activity 5 — find your sources (40 min, individual)

1. **Focus with AI:** brainstorm subtopics and keywords for your question (TECgpt / Gemini / Copilot).
2. **Search:** build one search string; run it in the Tec library and Google Scholar.
3. **Select at least 5 sources:** at least 3 peer-reviewed, at least 1 in English.
4. **Fill the literature matrix** — open every DOI.
5. **Validate with me** the sources you will use.

**Log your prompts** — Task 3 asks how you validated ideas and sources.

# Activity 6 — present your sources (graded, 5 %)

**3 minutes + 1 of feedback, each:**

- your topic, question and objective
- your 3–5 best sources — and why each is reliable
- one source you *rejected*, and why

Upload: `Matrícula_Actividad09.pptx`

**Rubric (Canvas)**

| | |
|---|---|
| Use of the concepts | 25 |
| Clarity | 25 |
| Coherence | 25 |
| Visual resources | 25 |

Feedback from the group: respectful and specific.

# Tasks 3 and 4 — independent work

**Task 3 — validate ideas and sources** (5 %)
Use AI to check the ideas in your sources; corroborate that every source exists and is credible; every claim backed by a reliable source.
`Matrícula_Validación_de_ideas.pdf`

**Task 4 — first draft with Agent Studio** (10 %)
MiTec → TECgpt → **Agent Studio** → **RigorIA**: structure, then section by section; you revise every part.
`Matrícula_Redacción_del_Articulo_IA.pdf`

**Start RigorIA like this:**

```
I am writing a [review / research /
brief] article on ____. The objective
is ____. Audience: ____.
Suggest a detailed structure with the
usual sections (Introduction, Methods,
Results, Discussion, Conclusions) and
suggested subsections.
```

Then: one section at a time, with length and content. Log every prompt.

Find it. Open it. Read it.

Then cite it.



lrodriguez15.github.io/ai-scientific-writing.html